

Lab report 5: EE-170L Computer Programming Lab Report

Title

The while Loop in C++

Objective

The objective of this lab is to learn and implement:

- The `while` loop
 - The nested `while` loop
 - The `do...while` loop
-

Theory

The while Loop

A `while` loop is a control flow statement that repeatedly executes a block of code as long as the given condition is true.

Syntax

```
while(condition)
{
    // Statements
}
```

Working

- The condition is checked first.
 - If the condition is true, the statements inside the loop execute.
 - After execution, the condition is checked again.
 - This process continues until the condition becomes false.
-

Nested while Loop

A nested `while` loop is simply a `while` loop inside another `while` loop.

Syntax

```
while(condition1)
```

```
{
    while(condition2)
    {
        // Statements
    }
}
```

Working

- The outer loop starts first.
 - For every iteration of the outer loop, the inner loop executes completely.
 - The process repeats until the outer loop condition becomes false.
-

The do...while Loop

The `do...while` loop is different from the `while` loop because it executes the loop body once before checking the condition.

Syntax

```
do
{
    // Statements
}
while(condition);
```

Working

- The loop body executes first.
 - Then the condition is checked.
 - If the condition is true, the loop repeats.
 - Otherwise, the loop terminates.
-

Program 1

Printing Numbers from 1 to 10 Using while Loop

```
#include<iostream>
using namespace std;

int main()
{
    int i = 1;

    // Yeh loop 1 se 10 tak numbers print karega
    while(i <= 10)
    {
```

```
        cout << i << endl;

        // Aglay number par janay ke liye
        i++;
    }

    return 0;
}
```

Output

```
1
2
3
4
5
6
7
8
9
10
```

Program 2

Nested while Loop

```
#include<iostream>
using namespace std;

int main()
{
    int i = 1;

    // Bahir wala loop rows ke liye hai
    while(i <= 3)
    {
        int j = 1;

        // Andar wala loop har row mein 1 se 5 tak numbers print karega
        while(j <= 5)
        {
            cout << j << " ";
            j++;

            // Aglay number ki taraf barho
        }

        cout << endl;

        // Agli row par janay ke liye
        i++;
    }

    return 0;
}
```

```
}
```

Output

```
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
```

Program 3

Printing Numbers from 1 to 10 Using do...while Loop

```
#include<iostream>
using namespace std;

int main()
{
    int i = 1;

    // Yeh loop kam az kam aik dafa zaroor chalega
    do
    {
        cout << i << endl;

        // Aglay number ke liye value barhao
        i++;

    }while(i <= 10);

    return 0;
}
```

Output

```
1
2
3
4
5
6
7
8
9
10
```

Conclusion

This experiment demonstrated the use of the `while`, nested `while`, and `do...while` loops in C++. The `while` loop executes only when its condition is true, whereas the `do...while` loop executes its body at least once before checking the condition. The nested `while` loop is useful for

solving problems that require repeated iterations within another loop, such as printing patterns or working with rows and columns.